MCO-1D METACHRONAL CASCADE ORNITHOPTER

PRE-FLIGHT CHECKLIST · Rev H · Software v4 · Confirm H_GEAR_MM before first flight

① BEFORE POWER-ON (walk-around)	ACTION
Airframe documents / registration	ON BOARD
Log — no outstanding maintenance squawks	CONFIRMED
Wing root pivot pins — locked and cotter pinned	CHECK
All 8 rope drives — seated in pulley grooves, taut	CHECK
Wing membranes — no tears or separation	CHECK
D-box CF skin — no delamination or damage	CHECK
Raked wingtips — bonded, no cracks at joint	CHECK
Gurney flaps — intact, not bent or detached	CHECK
VGs — strips bonded, none missing (drone)	CHECK
All 16 servo connectors — fully seated	SEATED
All 8 ODrive connectors — locked	LOCKED
CAN-FD termination resistors both ends	FITTED
Battery XT60 — fully mated, no corrosion	CHECK
Pitot tube — unobstructed, pointing fwd	CLEAR
Rangefinder lens — clean, unobstructed	CLEAR
GPS mast — secure, cable not strained	CHECK
All wiring — no chafing on moving parts	CHECK

② POWER-UP BIT (P-BIT) ~15 sec automated	ACTION
Connect battery — master switch ON	**ON**
RPi Zero 2W boots (green LED solid)	CONFIRM
P-BIT runs automatically on boot	
✓ Python env + firmware files present	PASS
✓ Wave library loads — VERSION displayed	PASS
✓ CAN-FD interface (can0) active	PASS
✓ All 8 ODrive nodes respond	PASS
✓ Pixhawk MAVLink heartbeat received	PASS
✓ pigpiod servo daemon running	PASS
✓ Log directory writable	PASS
■ **Any P-BIT FAIL → DO NOT PROCEED**	**RESOLVE**

③ SENSOR AND AVIONICS CHECK	ACTION
Pixhawk — solid green LED (ready to arm)	GREEN
Mission Planner — GPS 3D fix, HDOP < 2.2	CONFIRM
Airspeed at rest — < 2.0 m/s	CHECK
Battery voltage — > 22.0V (> 3.67V/cell)	CHECK
Rangefinder — distance to ground reading	READING
RC receiver — all 8 channels responsive	CHECK
Throttle — minimum, all switches safe	**MINIMUM**
Telemetry link to laptop — connected	CONNECTED

④ ENVIRONMENT	ACTION
Flying area clear — 30m radius minimum	CLEAR
Wind speed — < 8 m/s for first flights	CHECK
No precipitation, ice, or dense fog	CONFIRMED
Spotter present and briefed	READY
Emergency stop: throttle minimum → disarm	**BRIEFED**

⑤ TIER 2 SYSTEM EXERCISE ~90 sec (first flight of day)	ACTION
Run: python3 main.py (or --refly for 2nd+ flight)	**RUN**
✓ Battery volts/cell confirmed at start	PASS
✓ All 8 sweep servos: 60°→180°→210°→60°	PASS
✓ All 8 pitch servos: full range exercise	PASS
✓ Stow sequence W4→W3→W2→W1 at 20°/s	PASS
✓ Wings redeploy to 60° after stow test	PASS
✓ Motor spin 0.3 Hz for 2 sec (wings stowed)	PASS
✓ Motors idle — wings redeploy 60°	PASS
✓ Rangefinder height reading plausible	PASS
■ **Any Tier 2 FAIL → DO NOT ARM**	**RESOLVE**
--refly flag skips Tier 2 (same session)	

⑥ BEFORE ARMING	ACTION
Wave library mode — metachronal_90	SET
Sweep angles — all at 60° (deployed fwd)	CONFIRM
Throttle — minimum before arm	**MINIMUM**
■ **All personnel clear of wing arc (>1m)**	**CLEAR**
ARM on transmitter	**ARM**
✓ Pixhawk LED — solid green (armed)	GREEN

⑦ TAXI AND TAKEOFF	ACTION
Ground taxi — CH3 throttle, yaw for turns	ASYMM STROKE
Taxi amp: upstroke 80°, downstroke 4° max	
Advance throttle — wings bite at ~30%	SMOOTH
✓ Liftoff — amplitude auto-restores above 0.65m	AUTO
Climb — standard metachronal cascade 0.8 Hz	CLIMB
Optimiser CH6 — enable after 10m AGL	OPTIONAL

⑧ IN-FLIGHT LIMITS (MCO-1D)	ACTION
■ **Stall speed — 4.8 m/s (9.3 kt) all wings**	**NEVER BELOW**
Max cruise freq — 3.0 Hz sprint	DO NOT EXCEED
Max sweep angle — 210° (hard stop)	DO NOT FORCE
Min AGL for optimiser — 3.0 m	OBSERVE
CF-1 mode — J < 0.8 only, < 10 m/s	AUTO GATED
■ **Battery critical — 19.8V → LAND NOW**	**LAND NOW**
Low battery — 20.8V → plan landing	PLAN LAND
■ **Max pilot-commanded stoop — trained only**	**TRAINED ONLY**

⑨ LANDING AND PARK	ACTION
Approach — reduce throttle smoothly	SMOOTH
✓ Wings auto-limit amplitude below 0.65m AGL	AUTO
✓ Touchdown < 30mm AGL — motors auto-idle	AUTO
Disarm on transmitter — wings stow W4→W1	**DISARM**
Run Park Mode — python3 main.py park	RUN
Battery — remove and check voltage	CHECK
Log flight number and any squawks	LOG
■ **Any squawk — note before next flight**	**NOTE**

GENCY STOP: Throttle MIN → DI